Latu Kalita

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Research Interests

Theoretical and computational condensed matter physics, centred on the interplay of **topology**, **non-Hermiticity** and **periodic driving**. My current work spans Floquet engineering of higher- and hybrid-order topological phases, non-Bloch band theory and the non-Hermitian skin effect, and the entanglement structure of driven and disordered fermionic systems. More recently I have been working on Griffiths-type rare-region effects and the associated singular density of states in disordered Floquet chains. I am extending this towards infinite-randomness criticality, open quantum systems and field-theoretic descriptions of topological criticality.

Publications and Preprints

Published

[1] K. Roy, **L. Kalita**, B. Tanatar and S. Basu, *Floquet generation of hybrid-order topology and $\mathbb{Z}_2$-like bipolar localization*, Phys. Rev. Research **8**, 043094 (2026) [arXiv:2603.21954].

Preprints

[2] **L. Kalita**, K. Roy, B. Tanatar and S. Basu, *From traveling waves to synthetic gauge fields: A field-free route to superconducting diode effect*, arXiv:2610.29865 (2026).

Research Positions

Junior Research Fellow, School of Physics, IISER Thiruvananthapuram — *Oct. 2025 – present*
ANRF-funded project *Theoretical study of robust topological quantum systems and their characterization* · PI: Dr. Suraj S. Hegde

Education

M.Sc. in Physics · Indian Institute of Technology Guwahati — *2023 – 2025*
CGPA 8.99/10.0 · Thesis: *Non-Hermitian Topology and Floquet Skin Effect* · Advisor: Prof. Saurabh Basu

B.Sc. in Physics · Cotton University, Guwahati — *2020 – 2023*
CGPA 9.43/10.0 · First rank in the university · Gold Medal

Research Experience

IISER Thiruvananthapuram · Supervisor: Dr. Suraj S. Hegde — *Oct. 2025 – present*

- **Entanglement structure of driven and disordered Kitaev chains.** Numerical study of half-chain and topological entanglement entropy across disorder-induced transitions between Floquet Majorana phases; extraction of effective central charges and tests of infinite-randomness scaling.
- **Griffiths effects and rare regions in driven disordered chains.** Identified a Griffiths effect around the zero mode of the disordered Floquet Kitaev chain, traced to exponentially small rare-region level splittings, and tested whether an analogous effect arises at the π/T gap unique to the driven case; characterized through disorder-averaged densities of states and near-zero-energy wavefunction profiles.

Indian Institute of Technology Guwahati · Supervisor: Prof. Saurabh Basu — *July 2024 – July 2025*

- **Floquet hybrid-order topology and bipolar skin localization** [1]. Showed that periodic driving activates first-order helical edge states coexisting with higher-order corner modes in a Benalcazar–Bernevig–Hughes lattice, and identified a drive-controlled unipolar-to-bipolar skin localization transition. Characterized via a non-Bloch Floquet generalized-Brillouin-zone framework, real-space chiral winding numbers and spectral-localizer diagnostics.
- **Field-free superconducting diode effect from travelling-wave drives** [2]. Showed that a chemical potential driven in space and time breaks inversion and time-reversal symmetry without a magnetic field, generating a synthetic gauge flux in a helical Shiba chain. The resulting superconducting diode effect is tunable and reversible by the drive, reaching ∼50% rectification in the topological phase; analysed within a self-consistent Bogoliubov–de Gennes formalism.
- **M.Sc. thesis.** Skin effect and the breakdown of conventional bulk–boundary correspondence in the Su–Schrieffer–Heeger model and in two-dimensional p-wave superconductors, using generalized-Brillouin-zone construction and Floquet theory.

Cotton University, Guwahati · Supervisor: Dr. Mridula Baro — *2022 – 2023*

- Synthesis of biomass-derived porous carbon nanomaterials by hydrothermal carbonization of bamboo powder with $ZnCl_2$ activation, for energy-storage applications.

Talks, Schools and Workshops

- 66th DAE Symposium on Nuclear Physics, 2022 (participant).
- NPTEL certification, *Numerical Methods and Simulation Techniques for Scientists and Engineers*, IIT Guwahati, 2021.

Technical Skills

- **Numerical methods.** Exact diagonalization; self-consistent Bogoliubov–de Gennes solvers; Floquet and Magnus expansions, two-frame stroboscopic evolution; non-Bloch band theory and generalized-Brillouin-zone construction; real-space topological markers and spectral localizers; entanglement entropy, inverse participation ratio and level-statistics computation; transfer-matrix methods.
- **Programming.** Python (NumPy, SciPy, Matplotlib), C/C++, Fortran, Mathematica, Scilab.
- **Tools.** LaTeX

Fellowships and Awards

- **ANRF Junior Research Fellowship**, Anusandhan National Research Foundation, India — project *Theoretical study of robust topological quantum systems and their characterization* (2025 – present).
- **INSPIRE Scholarship**, Department of Science and Technology, India, for academic excellence in the basic sciences (2020 – 2025).
- **Gold Medal**, Cotton University — first rank, B.Sc. Physics (2023).

National Examinations

- **CSIR–UGC NET (Physics)**, December 2025 session — All India Rank 174 of 22,533. Qualified for the Junior Research Fellowship and Assistant Professor eligibility.
- **GATE (Physics)** 2025 — All India Rank 325.
- **IIT–JAM (Physics)** 2023 — All India Rank 271.

Teaching

- **Private tutor**, Physics — one-to-one tuition, 2026 – present. Weekly sessions electromagnetism, Mathematical physics, Quantum Mechanics;

Coursework and Languages

Selected graduate coursework. Quantum Field Theory · General Relativity · Quantum Information and Computation · Open Quantum System · Topics in Condensed Matter Physics · Advanced Statistical Mechanics · Solid State Physics · Mathematical Physics

Languages. English (professional, medium of instruction throughout) · Hindi (native) · Assamese (mother tongue)

Referees

Dr. Suraj S. Hegde — current supervisor
Assistant Professor, School of Physics, IISER Thiruvananthapuram, Kerala, India · surajhegde@iisertvm.ac.in

Prof. Saurabh Basu — M.Sc. project and thesis supervisor
Professor, Department of Physics, IIT Guwahati, Assam, India · saurabh@iitg.ac.in